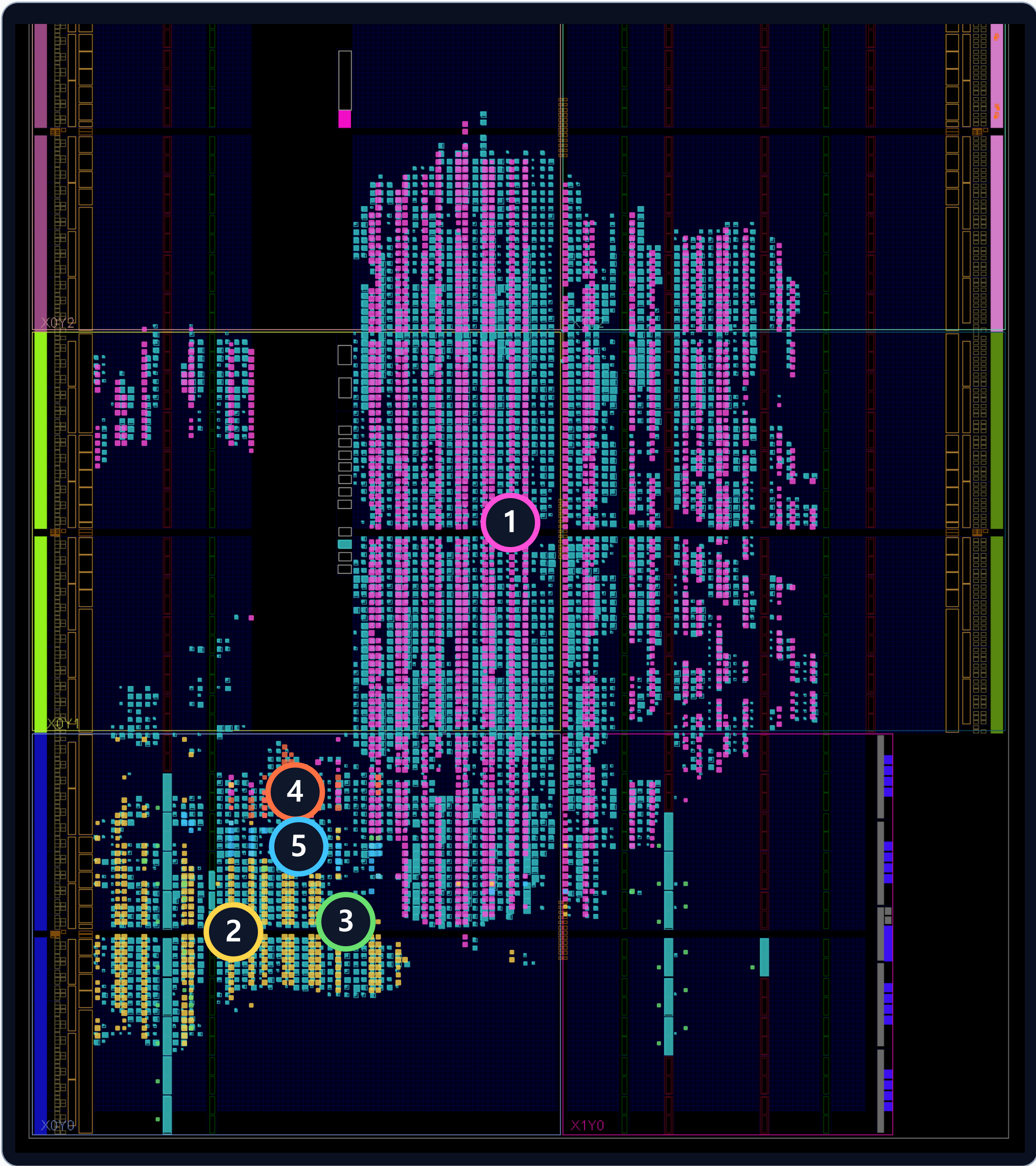
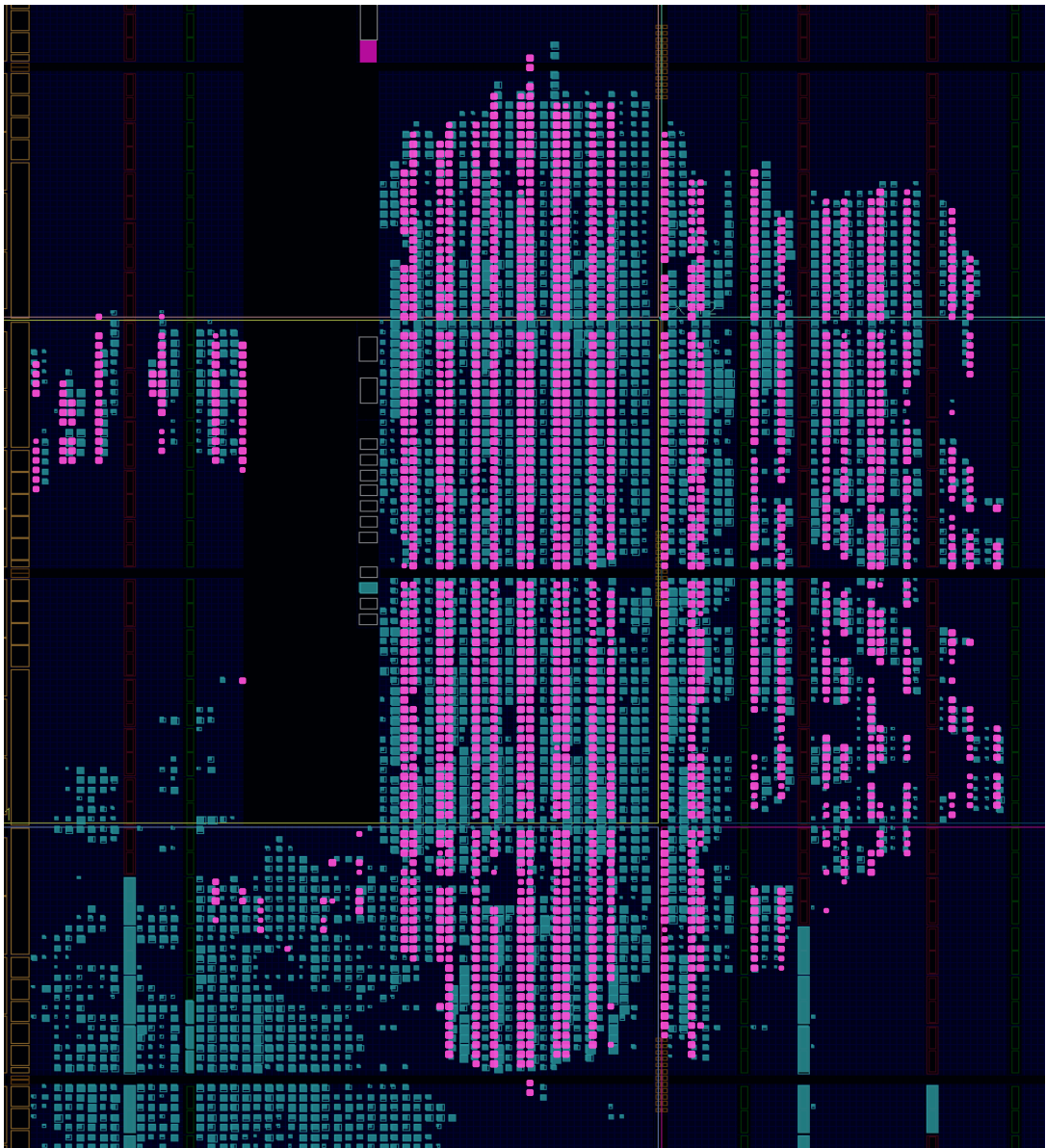


Post-route FPGA implementation and hierarchy placement

Vivado 2020.2 · Artix-7 XC7A100T · fixed routed checkpoint의 실제 Device View



(b) SNN accelerator에 속한 배치 셀만 분리 표시



- 1 SNN ECG accelerator
10,485 LUT / 6,652 FF
- 2 MicroBlaze processor
1,477 LUT / 1,212 FF / 3 DSP
- 3 Local memory
16 BRAM
- 4 Sample feeder
122 LUT / 176 FF
- 5 AXI / UART / interrupt
control and result transport

(c) 구현 범위와 정량 결과

signed 12-bit stream → Snapshot evidence → Final Membrane / WTA

Pure RTL accelerator

9,719 LUT · 5,038 FF · 0 BRAM · 0 DSP · WNS 8.184 ns

MicroBlaze integrated post-route system

12,494 LUT · 8,494 FF · 16 BRAM · 3 DSP · WNS 0.097 ns